

RESEARCH ARTICLE



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Uncertainty, Skewness, and the Business Cycle Through the MIDAS Lens

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ABSTRACT

We employ a mixed-frequency quantile regression approach to model the time-varying conditional distribution of the US real GDP growth rate. We show that monthly information on financial conditions improves the predictive power of an otherwise quarterly-only model. We combine selected quantiles of the estimated conditional distribution to produce novel measures of uncertainty and skewness. Embedding these measures in a VAR framework, we show that unexpected changes in uncertainty are associated with an increase in (left) skewness and a downturn in real activity. Business cycle effects are significantly downplayed if we consider a quarterly-only quantile regression model. We find the endogenous response of skewness to substantially amplify the recessionary effects of uncertainty shocks. Finally, we construct a monthly frequency version of our uncertainty measure and document the robustness of our findings.

Jel classification: C32, E32

1 | Introduction

The seminal contribution on “growth-at-risk” by Adrian, Boyarchenko, and Giannone (2019) has raised interest in the use of financial indicators for forecasting changes in the left tail of the distribution of real GDP growth.¹ Correctly quantifying skewness, a prime indicator of economic risk, is crucial for policymakers concerned with risk-management and, more in general, with the correct design of macroeconomic policies that aim at limiting business cycle costs due to rare but large shocks (see, e.g., Evans et al. 2015).

This paper shows that monthly data on financial conditions readily available to policymakers and researchers improve the forecast accuracy of the conditional distribution of real GDP growth in an otherwise quarterly data-only model. We augment Adrian et al.’s (2019) quantile regressions framework

with monthly realizations of the national financial conditions index (NFCI) (used at a quarterly level by Adrian et al., 2019) by implementing the unrestricted mixed-frequency approach (MIDAS) proposed by Foroni, Marcellino, and Schumacher (2015).² We document the superior out-of-sample predictive ability of our MIDAS model with respect to the quarterly data-only framework using the quantile combination approach proposed by Lima and Meng (2017) and via quantile score tests. Then, we construct measures of uncertainty and skewness by combining selected quantiles of the estimated conditional distribution of real GDP growth. Uncertainty is measured as the dispersion of such distribution, while we use a Kelley index for skewness. Both measures display substantial time dependence and feature abrupt variations in correspondence with extreme events such as the recessions of the early 1980s, the 2007–09 Great Recession, and the COVID-19 pandemic one. Our novel measures are shown

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to correlate with measures of real uncertainty and skewness such as the macroeconomic uncertainty proxies estimated by Jurado, Ludvigson, and Ng (2015), Carriero, Clark, and Marcellino (2018), Shin and Zhong (2020), and Ludvigson, Ma, and Ng (2021) and the skewness measures proposed by Rossi and Sekhposyan (2015), Iseringhausen, Petrella, and Theodoridis (2023), and Salgado, Guvenen, and Bloom (2023).

Endowed with our novel uncertainty and skewness measures, we conduct a VAR analysis that models a battery of standard indicators of the business cycle and our measures of uncertainty and skewness. Following an unexpected increase in uncertainty, our VAR points to a statistically significant and quantitatively relevant downturn in real activity. We find that exogenous changes in uncertainty are responsible for 8% of the forecast error variance of real GDP at a 2-year horizon, 18% for that of the unemployment rate, and 10% and 12% for that of investment and hours, respectively. Repeating the same exercise with the uncertainty and skewness measures estimated with quarterly data only, we find the impact of changes in uncertainty on the business cycle to be much lower, that is, 3% of the forecast error of real GDP and investment and 7% of that of unemployment and hours. We attribute this discrepancy to the fact that when working with quarterly data, relevant spikes in the time series of unexpected changes in uncertainty are overlooked, something that does not happen when we put our mixed-frequency model at work.

A novel result of our VAR empirical analysis is the response of skewness, i.e., an uncertainty shock is associated with left skewness (i.e., lower quantiles responding more than upper ones). Is this endogenous response of skewness relevant for the quantification of the business cycle effects of uncertainty shocks? We address this question by running a counterfactual exercise that shuts down the endogenous response of skewness to an exogenous variation of uncertainty. We find that muting the response of skewness—i.e., freezing the indirect impact of uncertainty shocks on real activity due to the response of skewness in the baseline model—implies a substantially milder drop in real activity. For instance, the peak response of real GDP is twice as large in the data as in a counterfactual world in which skewness stays still after an uncertainty shock. Hence, we estimate the “uncertainty-skewness multiplier” to be as large as 2.

The final part of our paper turns our analysis to a monthly frequency one. We do so by considering a version of our model where we exploit the monthly information on financial conditions as it becomes available *within* the predictive quarter of interest.³ Via this exercise, we produce novel measures of uncertainty and skewness at a monthly frequency. We then show that such measures are (i) highly correlated with our quarterly measures, (ii) generate similar results in terms of impulse responses of real activity indicators and role played by endogenous skewness (as a magnifier of the business cycle effects of uncertainty shocks) when we run a VAR exercise similar to the one described above, and (iii) point to the relevance of uncertainty shocks for the business cycle comparable to the one documented with the same data and empirical model but difference proxies for uncertainty, which we borrow from the literature (namely, Jurado, Ludvigson, and Ng 2015; Ludvigson, Ma, and Ng 2021; Carriero, Clark, and Marcellino 2018; Shin and Zhong 2020).

Our results point to the importance of the following: (i) considering monthly frequency information related to the state of financial markets to predict the left tail of the conditional density of the quarterly growth rate of real GDP and compute measures of uncertainty and skewness, (ii) modeling skewness in VAR investigations to take into account not only the direct effects that uncertainty can have on real activity but also the indirect effects via the endogenous response of the left tail of the real GDP density, and (iii) building up theoretical frameworks that can help us understand how to design optimal policies to tackle uncertainty shocks in the presence of endogenous skewness.

The remainder of the paper is structured as follows. Section 2 draws connections with the existing uncertainty-related literature.⁴ Section 3 documents the performance of our mixed-frequency approach in predicting the conditional density of real GDP growth. Section 4 proposes novel measures of uncertainty and skewness based on selected quantiles of the estimated time-varying conditional density. Section 5 presents VAR investigations involving the above-mentioned novel measures and a battery of macroeconomic indicators. Section 6 deals with the construction and employment in a VAR context of our monthly measures of uncertainty and skewness. Section 7 concludes the study.

2 | Connections With the Literature

The closest contributions to ours are Rossi and Sekhposyan (2015), Segal, Shaliastovich, and Yaron (2015), and Forni, Gambetti, and Sala (2021a). Rossi and Sekhposyan (2015) construct an index of uncertainty based on how unexpected the forecast error associated with the realization of real GDP growth is relative to the unconditional forecast error distribution, where the forecasts are those of the Survey of Professional Forecasters.⁵ They also distinguish between news that are unexpectedly positive vs. negative and construct measures of upside and downside uncertainty. Running a VAR analysis, they find the latter to have a substantially larger effect on real activity than the former. Segal, Shaliastovich, and Yaron (2015) decompose aggregate uncertainty into “good” and “bad” volatility components—respectively associated with positive and negative innovations to macroeconomic growth, such as the high-tech revolution of early-mid 1990s and the collapse of Lehman Brothers in September 2008. They document an asymmetric effect of these two types of uncertainty on the real and financial cycles, good uncertainty being associated with positive innovations to real GDP, consumption, and investment vs. bad uncertainty being recessionary. Forni, Gambetti, and Sala (2021a) combine quantile regressions and structural VARs to model the conditional density of real GDP growth and identify its determinants. They find that the recessionary effects often attributed to the second-moment shocks are actually driven by shocks to the lower tail of such a conditional density, i.e., downside uncertainty. Differently, shocks to the higher tail (upside uncertainty) are mildly expansionary. With respect to these three papers, we (i) deal with a mixed-frequency framework and show that monthly information on financial conditions is important to predict the stance of the business cycle in a nonlinear context, (ii) develop novel measures of uncertainty and skewness at quarterly and

monthly frequencies, and (iii) show that accounting for the endogenous reaction of skewness to exogenous changes in uncertainty is crucial to correctly quantify the business cycle effects of uncertainty shocks.

Adrian, Boyarchenko, and Giannone (2019) model the evolution of the conditional density of real GDP growth and show that financial conditions—captured by the national financial conditions index (NFCI) produced by the Federal Reserve Bank of Chicago—are crucial to predict “growth-at-risk”, i.e., the evolution of the lower tail of the predicted density.⁶ They do so by employing quantile regressions at quarterly frequencies. Our paper shows that mixing frequencies and employing monthly realizations with a parsimonious modeling approach similar to the one proposed by Adrian, Boyarchenko, and Giannone (2019) improves the forecasting performance of the quantile regression model, above all as far as skewness (“growth-at-risk”) is concerned. Iseringhausen, Petrella, and Theodoridis (2023) propose a data-rich approach to measure expected macroeconomic skewness in the U.S. economy. In line with our results, they find expected macroeconomic skewness to be procyclical. Ghysels (2014) and Aastveit, Foroni, and Ravazzolo (2017) find that high-frequency information can significantly improve the estimation of density forecasts of macroeconomic variables. Our empirical results point in particular to the relevance of employing monthly data on financial conditions to estimate the conditional density of the growth rate of real GDP. In this sense, our paper connects to the ones of Giglio, Kelly, and Pruitt (2016), Alessandri and Mumtaz (2017), González-Rivera, Maldonado, and Ruiz (2019), Delle Monache, De Polis, and Petrella (2021), Adams et al. (2021), and Galvão and Owyang (2022) that show that financial conditions (or factors correlated with them) are important to predict macroeconomic risk/vulnerable growth.⁷ Our paper also complements contributions that have stressed the interaction between financial conditions and tail events (Caldara, Scotti, and Zong 2021), possibly generated by first-moment shocks. Our contribution focuses on the role that higher frequency information can play with respect to a quarterly-only model.

Other departures with respect to the framework proposed by Adrian, Boyarchenko, and Giannone (2019) have been investigated by the literature, i.e., nonlinearities and/or a panel data approach. Clark et al. (2021) cover both such dimensions, while Plagborg-Müller et al. (2020) and Reichlin, Ricco, and Hasenzagl (2020) investigate the potentially nonlinear role played by financial indicators in predicting real GDP growth. Our paper connects with the contributions that have explored the state-dependent effects of uncertainty shocks, typically finding them to be larger during recessions (Caggiano, Castelnuovo, and Groshenny 2014; Caggiano, Castelnuovo, and Nodari 2022; Pellegrino, Castelnuovo, and Caggiano 2023; Andreasen et al. 2024), possibly due to financial frictions (Alessandri and Mumtaz 2019; Alessandri and Bottero 2020; Alfaro, Bloom, and Lin 2024).

3 | Real GDP Growth's Conditional Density: A MIDAS Approach

Let y_{t+h} be the annualized average growth rate of GDP between t and $t+h$ (i.e., $y_{t+h} = \frac{400}{h} \sum_{i=1}^h \Delta \log(GDP_{t+i})$, where Δ is the first-difference operator) and x_t the vector that contains all the conditioning variables, including a constant. As shown by Koenker and

Bassett (1978), the $100(\tau)$ -th conditional quantile of y_{t+h} can be estimated by solving the following minimization problem:

$$\hat{\beta}_\tau = \underset{\beta_\tau}{\operatorname{argmin}} \sum_{t=1}^{T-h} \left(\tau 1_{(y_{t+h} \geq x_t \beta_\tau)} |y_{t+h} - x_t \beta_\tau| + (1-\tau) 1_{(y_{t+h} \leq x_t \beta_\tau)} |y_{t+h} - x_t \beta_\tau| \right) \quad (1)$$

where $\tau \in (0,1)$ and 1_0 denotes an indicator variable which takes value “1” if the argument is true and “0” otherwise. The time-varying fitted values of (1), i.e., $\hat{Q}_\tau(y_{t+h}|x_t) = x_t \hat{\beta}_\tau$, are consistent estimators of the $100(\tau)$ -th quantile of the objective distribution, given x_t . The quarterly data-only approach by Adrian, Boyarchenko, and Giannone (2019) can be replicated by setting $x_t = \{1, y_t, NFCI_t\}$. As anticipated above, the alternative we propose is to use an (unrestricted) MIDAS model that features both a lagged realization of the quarterly growth rate and three monthly realizations of NFCI as covariates. Formally, the MIDAS specification features the following set of predictors: $x_t^* = \{1, y_t, NFCI_{t,m3}, NFCI_{t,m2}, NFCI_{t,m1}\}$.⁸ Model (1) can be estimated directly without restrictions on the parameters by replacing x_t with x_t^* .⁹ We consider the sample 1971Q1–2023Q1. Amburgey and McCracken (2023b) and Amburgey and McCracken (2023a) show that quantile regressions are sensitive to extreme outliers such as those observed during COVID-19. Hence, we control for the pandemic-related extreme volatility of real GDP growth at the end of the sample by applying the scaling strategy proposed by Lenza and Primiceri (2022).¹⁰ Further details and evidence are provided in our Appendix (Figure S1). Our results are robust to dropping the pandemic observations from the investigated sample (see Figure S2 in the Appendix and Castelnuovo and Mori (2022)).

The following Sections 3.1–3.3 document the following: (i) the nonlinear relationship between financial conditions and different quantiles of the conditional density of real GDP growth, (ii) the out-of-sample models' point-forecast accuracy in terms of root mean square error (RMSE) and mean absolute error (MAE), (iii) the out-of-sample ability of the model to capture downside risk (using the quantile score on lower quantiles and the quantile-weighted continuous ranked probability score), and (iv) the evolution of the conditional quantiles delivered by the two models and their ability to timely pick up the arrival of recessionary periods.¹¹ We focus our attention on a short-term forecast and consider $h = 1$.¹²

3.1 | Nonlinearity

Figure 1 documents the estimated quantile regression coefficients for different quantiles of our MIDAS model and for the quarterly-only model. Quantile coefficient estimates that fall outside the confidence bounds point to a nonlinear relation between GDP growth and the predictive variable. Our evidence confirms the finding by Adrian, Boyarchenko, and Giannone (2019), who document the significant difference of the estimated slopes for the NFCI at both the lower and the upper quantiles with respect to the OLS ones and the stability of those of real GDP growth with a sample ending in 2015Q4. Importantly, the same observation holds true for the three estimated slopes for the NFCI in our MIDAS framework. Moreover, we detect a switch in sign when moving from the first to the second monthly realization of the NFCI, with the

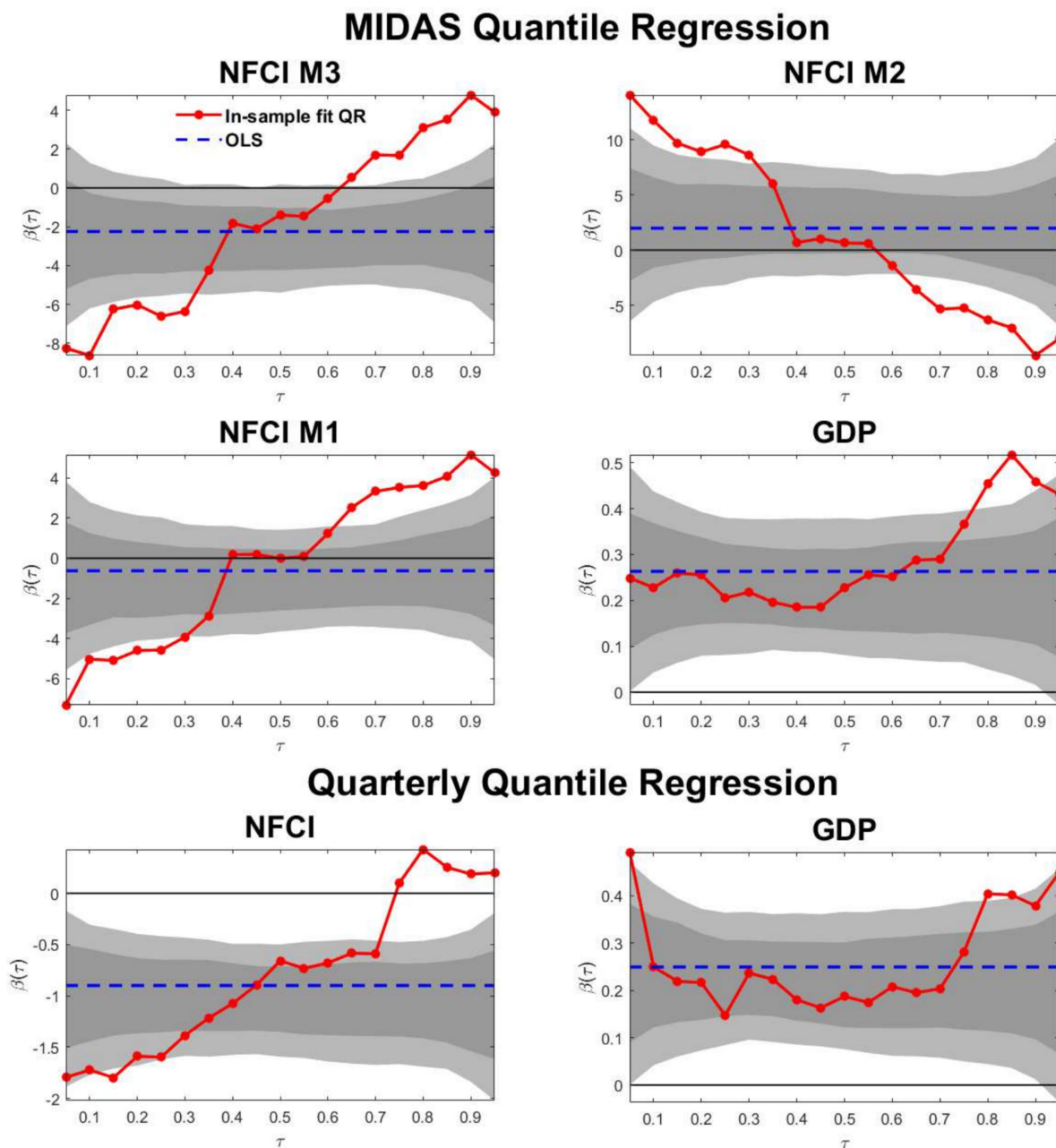


FIGURE 1 | MIDAS vs. quarterly-only models: Estimated quantile regression coefficients for different quantiles. The figure reports the estimated coefficients in quantile regressions of one quarter ahead real GDP growth on current real GDP growth and NFCI. MIDAS model in upper panels; quarterly-only model in lower panels. Following Adrian, Boyarchenko, and Giannone (2019), we report the 68% and 90% confidence bounds for the null hypothesis that the true data-generating process is a general, flexible linear model for growth and financial conditions (VAR with 4 lags); bounds are computed using 2000 bootstrapped samples. Quantile coefficient estimates that fall outside this confidence bound point to a nonlinear relation between GDP growth and the predictive variable. Sample: 1971Q1–2023Q1.

lower (upper) quantiles of the conditional density of real GDP growth being characterized by a negative (positive) impact of the NFCI during the first month of the quarter, while the opposite is recorded for the second month. We also observe a switch in a such sign when moving from the second one to the third one. This sign heterogeneity suggests the presence of a “dynamic correction” associated with financial conditions when it comes to predicting real GDP growth. Intuitively, our MIDAS model tends to predict more pronounced deteriorations in the left-tail of the distribution when there is a rise

in financial stress within the quarter (and especially in the last month of the quarter) as captured by monthly NFCI realizations. The value of this monthly information is obviously bound not to emerge when using quarterly data only. An illustrative example of the potential relevance of this mechanism is the advent of the COVID-19 pandemic in March 2020. As shown in our Appendix (Figure S4), the increase in the quarterly NFCI in 2020Q1 is relatively moderate. However, monthly financial conditions change substantially within such a quarter, with the index remaining relatively stable in

January and February but displaying a sharp spike in March.¹³ The MIDAS model does account for the monthly evolution of financial conditions and, therefore, provides a more precise forecast on the downside risk associated with the following quarter's realization of the growth rate of real GDP.

3.2 | Evolution of Conditional Densities

Figure 2 shows the (in-sample) evolution of selected conditional quantiles delivered by the two models. The general pattern appears to be the same in terms of median value and quantile dispersion, with the lower quantile of the distributions moving much more than the upper quantile. A clear difference arises when comparing the relative ability of the two models to pick up recessions, with the conditional density implied by the MIDAS model adjusting more timely than the one predicted by the model exploiting the information carried by the NFCI at a quarterly frequency only. The actual realization of the real GDP growth rate is below the 10-th quantile of the quarterly-only model's conditional density for a variety of recessions, including

the 2007-09 global financial crisis and the COVID-19 recession. Differently, the MIDAS model captures recessions in general (including these two extreme events), although it fails to fully pick up (very much like the quarterly model) the downturn in the early 1990s. Our Appendix (Figure S2) shows that the evolution of the estimated quantiles is robust to excluding COVID-19 observations.

3.3 | Predictive Power

We evaluate the out-of-sample predictive power of the competing models at hand in terms of both point-forecast performance and ability to capture tail risk. Following Adrian, Boyarchenko, and Giannone (2019), we work with a recursive forecasting scheme, using as initial estimation sample 1971Q1-1989Q4 (the out-of-sample forecast evaluation sample ends in 2023Q1). First, we assess the point-forecast ability of the competing models. As stressed by Carriero, Clark, and Marcellino (2020), even if quantile regression models are not explicitly designed to obtain point estimates, it can be useful to know how such models perform as far as the

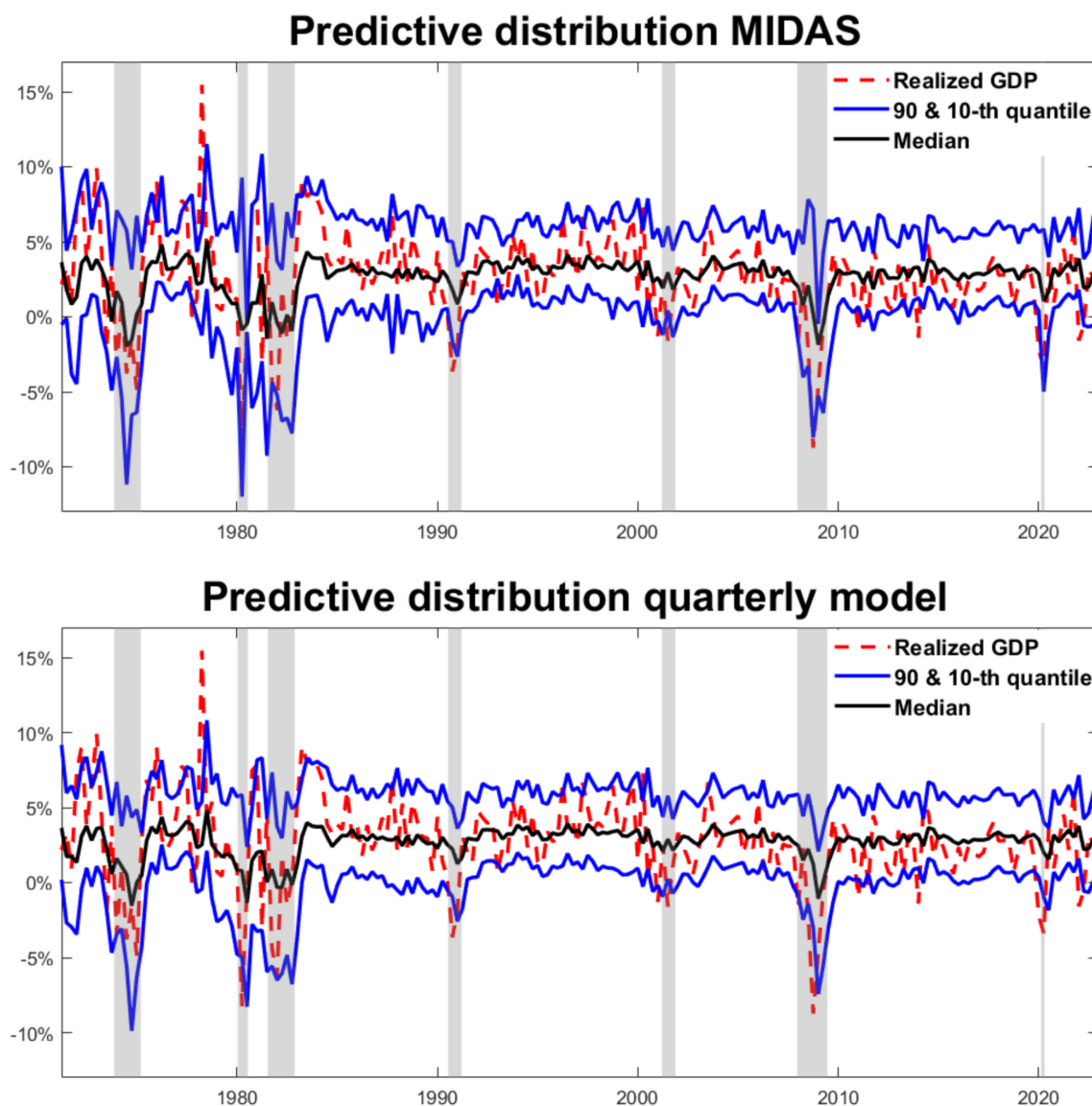


FIGURE 2 | MIDAS vs. quarterly-only models: Predictive densities of the growth rate of real GDP. Forecasting horizon: One quarter.

central tendency of the distribution is concerned. As in Lima and Meng (2017), we approximate the mean of the one-step ahead real GDP growth's conditional density by adequately combining its conditional quantiles (an approach termed “quantile combination approach”). Formally, the one-quarter ahead point forecast implied by our quantile regressions is computed as follows:

$$\hat{y}_{t+1} = \sum_{\tau \in J} \frac{1}{n} \hat{Q}_{\tau}(y_{t+1} | x_t) \quad (2)$$

where n is the length of J and J goes from 0.05 to 0.95 with steps of size 0.05—i.e., $J = \{0.05, 0.1, 0.15, \dots, 0.95\}$ —see Carriero, Clark, and Marcellino (2022) for this selection of quantiles. We then use the so-obtained point forecasts to compare the relative out-of-sample forecasting performance of the quarterly-only vs. MIDAS frameworks in a standard fashion.

Table 1 documents the outcome of this exercise. It reports the ratios of the relative RMSE and MAE of the MIDAS model over the quarterly-only model. Both the RMSE and the MAE point to the MIDAS model as the better-performing one. Reassuringly, this indication is also supported by the Diebold–Mariano test, which points to the statistical superiority (at the 5% level) of the MIDAS model vs. the quarterly-only competitor.

Another test we conduct is the quantile score, which is based on the assessment of the models' ability to capture tail risk (Giacomini and Komunjer 2005). We evaluate the relative out-of-sample accuracy of the 5-*th* and 10-*th* conditional quantiles. While the 5-*th* quantile is a standard benchmark for these tests, the choice of the 10-*th* quantile is motivated by the fact that this is the lower quantile of interest that we will use to construct our novel uncertainty and skewness measures. Conditional on these quantiles, we compute the quantile score as follows:

TABLE 1 | Point-forecast accuracy.

	MIDAS model/quarterly model
RMSE	0.98** (0.01)
MAE	0.98** (0.01)

Note: Ratios of root mean square errors and mean average errors computed by considering a recursive forecasting exercise. Initial estimation sample: 1971Q1–1989Q4. Out-of-sample period: 1990Q1–2023Q1. Values lower than one point to the superiority of the MIDAS model in a predictive sense. The differentials in the loss functions are corrected for heteroskedasticity and autocorrelation using Newey and West (1987). *p*-values of the Diebold and Mariano (1995) test in brackets. The test is conducted on a one-sided basis, such that the alternative hypothesis is that the MIDAS model is more accurate than the quarterly only.

TABLE 2 | Density accuracy: MIDAS vs. quarterly-only model.

	Out-of-sample	Out-of-sample rec.	In-sample	In-sample rec.
$\overline{QS}_{MIDAS}^{0.05} / \overline{QS}_Q^{0.05}$	0.96 (0.12)	0.83* (0.10)	0.87** (0.05)	0.56* (0.10)
$\overline{QS}_{MIDAS}^{0.1} / \overline{QS}_Q^{0.1}$	0.94 (0.14)	0.73* (0.07)	0.91 (0.11)	0.67* (0.07)
$\overline{qwCRPS}_{MIDAS} / \overline{qwCRPS}_Q$	0.97 (0.11)	0.88* (0.06)	0.96* (0.09)	0.83* (0.06)

Note: Ratios of mean quantile scores for 5-*th* and 10-*th* quantiles and quantile-weighted continuous ranked probability scores - both in and out-of-sample (estimation sample 1971Q1–1989Q4, and out-of-sample period 1990Q1–2023Q1) and for all observations vs. for recessionary periods only (NBER based recession indicator). Values lower than one point to the superiority of the MIDAS model. The differentials in the (quadratic) loss functions are corrected for heteroskedasticity and autocorrelation using Newey and West (1987). *p*-values of the Diebold and Mariano (1995) test in brackets. The test is conducted on a one-sided basis, such that the alternative hypothesis is that the MIDAS model is more accurate than the quarterly only. Number of recessionary periods: 29 in-sample; 14 out-of-sample.

$$QS_{t+1}^{\tau} = 2 \left(y_{t+1} - \hat{Q}_{\tau}(y_{t+1}) \right) \left(\tau - 1_{(y_{t+1} \leq \hat{Q}_{\tau}(y_{t+1}))} \right)$$

where $1_{(y_{t+1} \leq \hat{Q}_{\tau}(y_{t+1}))}$ has a value of 1 if the GDP growth realization is at or below the τ -*th* quantile and 0 otherwise.

The advantage of using the quantile score is that it is associated with an asymmetric loss function. Hence, the assessment of the accuracy of the estimated quantiles assigns a larger weight to overpredictions than to underpredictions. This implies that missing deep recessions is more costly (according to this loss function) than missing strong expansions, an assessment in line with the risk management approach undertaken by policymakers (Evans et al. 2015). In addition, the out-of-sample quality of the predictive density can be tested via the quantile-weighted continuous ranked probability score (qwRPS) proposed by Gneiting and Ranjan (2011), which is a weighted sum of quantile scores at a range of quantiles with an asymmetric scoring function that gives more weight to the left tail quantiles. The quantile-weighted continuous ranked probability score is defined as follows:

$$qwCRPS_{t+1} = \frac{1}{n} \sum_{\tau \in J} v(\tau) QS_{t+1}^{\tau}$$

where $v(\tau) = (1 - \tau)^2$, n is the length of J , and $J = \{0.05, 0.1, 0.15, \dots, 0.95\}$.

Table 2 reports the results of the above-mentioned tests in terms of out-of-sample and in-sample forecasting exercises as well as conditional on recessions only in the out-of-sample and in-sample versions. Overall, the MIDAS model is relatively more accurate in capturing tail risk out-of-sample. We find that $\overline{QS}_{MIDAS}^{0.05} / \overline{QS}_Q^{0.05} = 0.96$ and $\overline{qwCRPS}_{MIDAS} / \overline{qwCRPS}_Q = 0.97$ (*p*-values are 0.12 and 0.11).¹⁴ Similar results are obtained considering the 10-*th* quantile when computing the quantile score. The benefits of employing the MIDAS model become particularly evident when predicting (NBER-dated) recessionary periods. For such episodes, the quantile score ratios stand at 0.83, 0.73, and 0.88 for the 5-*th*, 10-*th*, and weighted measure (*p*-values are 0.1, 0.07, and 0.06).¹⁵ Our evidence points to a competitive performance of the MIDAS model when compared to the one of the quarterly-only model both in terms of point forecast and in the ability to capture tail risk. Violations of the lower quantiles are found to be smaller in magnitude than those implied by the quarterly-only framework (QS) and the overall quality of the predictive densities increases (*qwCRPS*).

All in all, the empirical evidence provided above points to our MIDAS approach as a competitive one with respect to the quarterly data-only framework.

4 | Macroeconomic Uncertainty and Skewness Measures

We derive measures of macroeconomic uncertainty and skewness by, respectively, quantifying the time-varying dispersion and asymmetry of the conditional density of real GDP growth estimated with our MIDAS framework.

4.1 | Uncertainty

Uncertainty at time t is defined as follows:

$$UNC_t = \hat{Q}_{0.9}(y_{t+1}|x_t) - \hat{Q}_{0.1}(y_{t+1}|x_t) \quad (3)$$

The logic behind this characterization of uncertainty is that conditional on our predictive analysis, uncertainty can be proxied by the spread between selected quantiles of the conditional distribution of output growth. The intuition is the following. In periods of high uncertainty, agents' expectations—which in our case are expressed in the form of a complete probability assessment over the

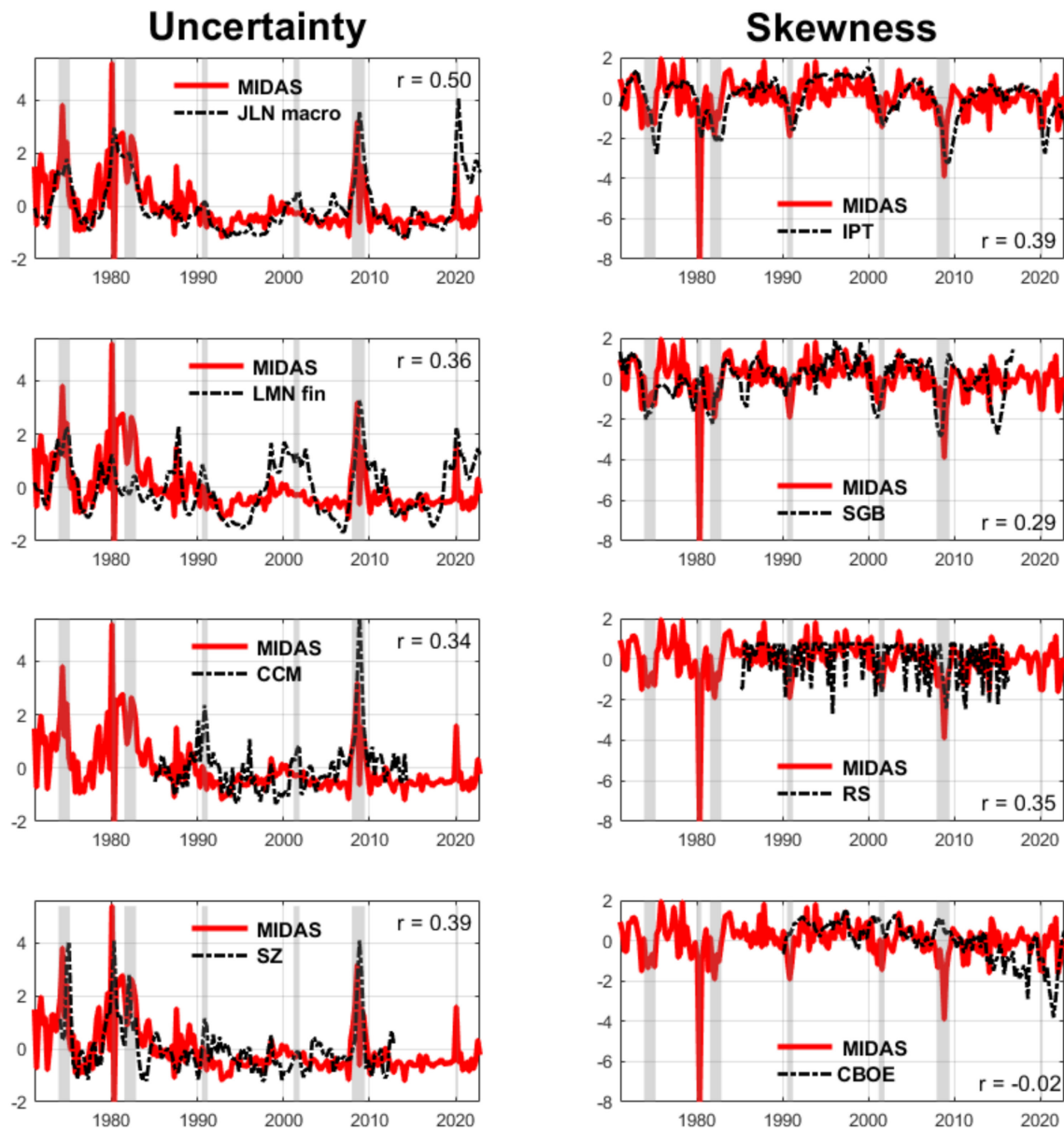


FIGURE 3 | Uncertainty and skewness measures estimated with the MIDAS model: Comparison with existing indices. Uncertainty: Macro uncertainty ($h = 1$ quarter) of Jurado, Ludvigson, and Ng (2015) (JLN macro); financial uncertainty ($h = 1$ quarter) of Ludvigson, Ma, and Ng (2021) (LMN fin); macroeconomic uncertainty of Carriero, Clark, and Marcellino (2019) (CCM); and Shin and Zhong (2020) (volatility of industrial production's residuals) (SZ). Skewness: Macro skewness of Iseringhausen, Petrella, and Theodoridis (2023) (IPT); Salgado, Guvenen, and Bloom (2023) (based on companies' sales growth rate) (SGB); Rossi and Sekhposyan (2015) (downside uncertainty at $h = 0$) (RS); and CBOE skewness index (from S&P500 out-of-the-money options) (CBOE). Indices are standardized for comparison. Correlations are computed based on the longest possible sample given data availability per each considered pair.

future realization of GDP growth—are dispersed. Hence, agents assign a positive probability to the occurrence of a wide set of possible states of the world. Conversely, in periods of low uncertainty, the probability of events on the tail of the conditional density is lower. The choice of the 0.9/0.1 quantiles follows the one in Forni, Gambetti, and Sala (2021a) and Salgado, Guvenen, and Bloom (2023).

Figure 3 (left column) plots our uncertainty measure constructed as in (3) and based on the full sample as in, e.g., Jurado, Ludvigson, and Ng (2015) and compares it with other proxies proposed by the literature. Our novel measure is clearly countercyclical, with peaks during recessions. The highest correlation (correlation coefficient: 0.50) is the one with the macroeconomic uncertainty series proposed by Jurado, Ludvigson, and Ng (2015), an evidence that suggests that our measure is likely to capture macroeconomic uncertainty (as opposed to, say, financial uncertainty). This is not surprising, given that our measure of uncertainty is based on time-varying quantiles of real GDP growth's conditional density. A positive correlation is also recorded with the macroeconomic uncertainty measures by Carriero, Clark,

and Marcellino (2018) and Shin and Zhong (2020) and the financial one proposed by Ludvigson, Ma, and Ng (2021).¹⁶

4.2 | Skewness

The estimated conditional density of real GDP growth can also be exploited to construct a measure of macroeconomic skewness. Following Salgado, Guvenen, and Bloom (2023), the indicator of skewness is based on the normalized Kelley index (Kelley 1974), which is defined as follows:

$$SKEW_t = \frac{\hat{Q}_{0.9}(y_{t+1}|x_t) - \hat{Q}_{0.5}(y_{t+1}|x_t)}{\underbrace{\hat{Q}_{0.9}(y_{t+1}|x_t) - \hat{Q}_{0.1}(y_{t+1}|x_t)}_{\text{Right tail share}}} - \frac{\hat{Q}_{0.5}(y_{t+1}|x_t) - \hat{Q}_{0.1}(y_{t+1}|x_t)}{\underbrace{\hat{Q}_{0.9}(y_{t+1}|x_t) - \hat{Q}_{0.1}(y_{t+1}|x_t)}_{\text{Left tail share}}} \quad (4)$$

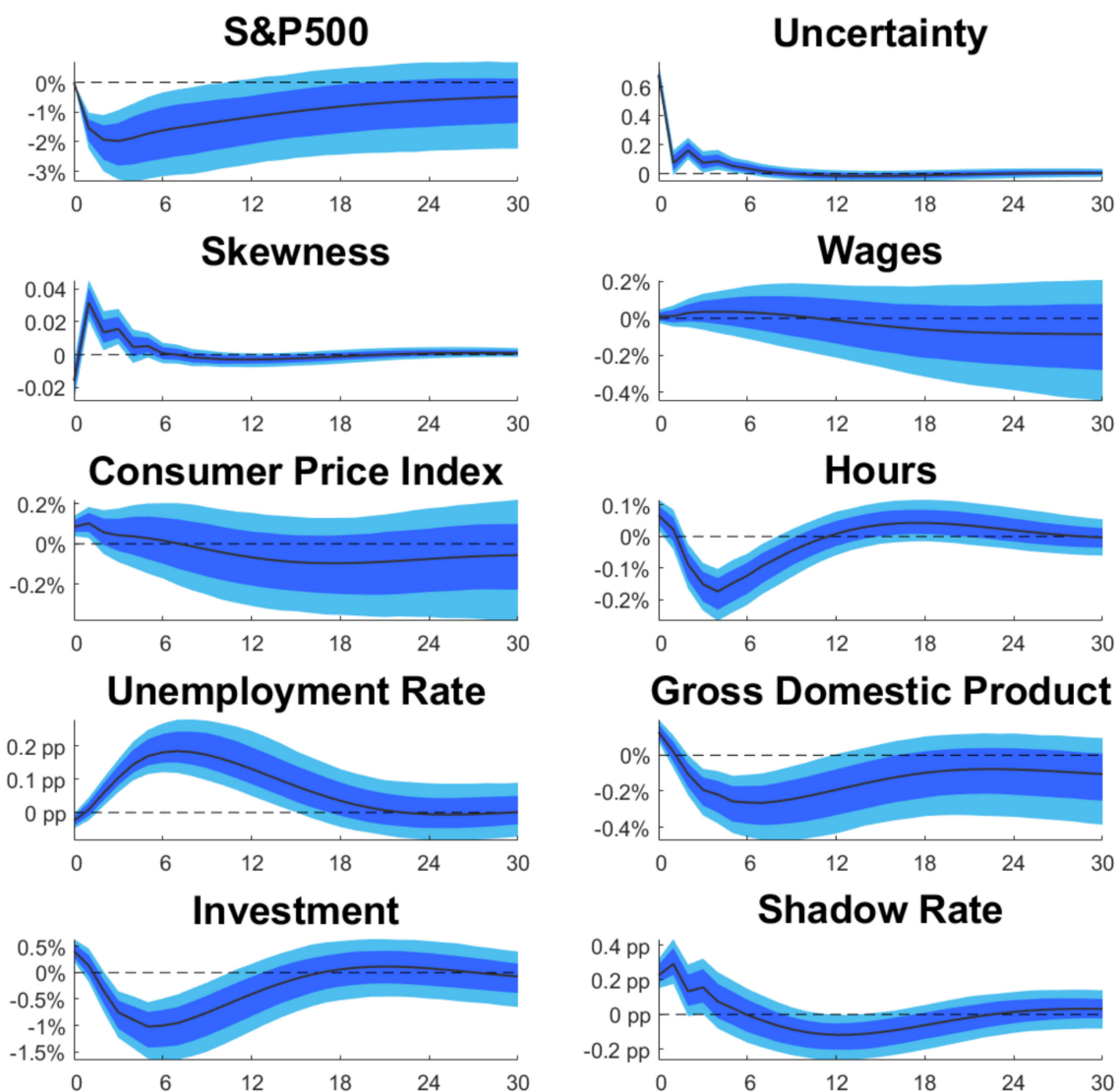


FIGURE 4 | Impulse responses to an unexpected change in uncertainty: uncertainty and skewness measures computed with the MIDAS model. VAR model featuring 4 lags. Shaded regions: 68% and 90% posterior credible sets. Unit variance shocks.

This measure offers a straightforward decomposition of the share of total dispersion that is attributable to the left and the right tails of the time-varying conditional distribution. Positive values indicate that the right tail is accountable for more than one-half of the total dispersion, so that the distribution is right skewed. Conversely, negative values of the Kelley index signal vulnerable growth (left skewness).

The measure of skewness obtained using (4) and the MIDAS model is plotted in Figure 3 (right column). The main indication that arises is that also macroeconomic skewness fluctuates substantially over time, and distributions become particularly left-skewed right before or during recessions. It is of interest to compare our measure of skewness with the one proposed by Iseringhausen, Petrella, and Theodoridis (2023) (which is constructed via a data-rich approach that models a variety of real activity indicators), the one based on US companies' sales growth

rate proposed by Salgado, Guvenen, and Bloom (2023), the one by Rossi and Sekhposyan (2015) (which measures uncertainty associated with news or outcomes that are unexpectedly negative, e.g., lower GDP than expected), and the CBOE skewness index (which measures the perceived tail risk of the distribution of S&P500 returns over a 30-day horizon). Our skewness index correlates positively with the ones by Iseringhausen, Petrella, and Theodoridis (2023) and Rossi and Sekhposyan (2015) (respectively, 0.39 and 0.35), something that confirms that our novel measures are macro-based measures (as opposed to financial ones). As in the case of uncertainty, such high(er) correlations are probably due to the fact that, as in the two papers mentioned above, we are modeling the conditional quantiles of real GDP growth (as opposed to financial or micro data). Our skewness measure correlates basically zero with the CBOE one, while the correlation between ours and Salgado et al.'s (2023) is 0.29.

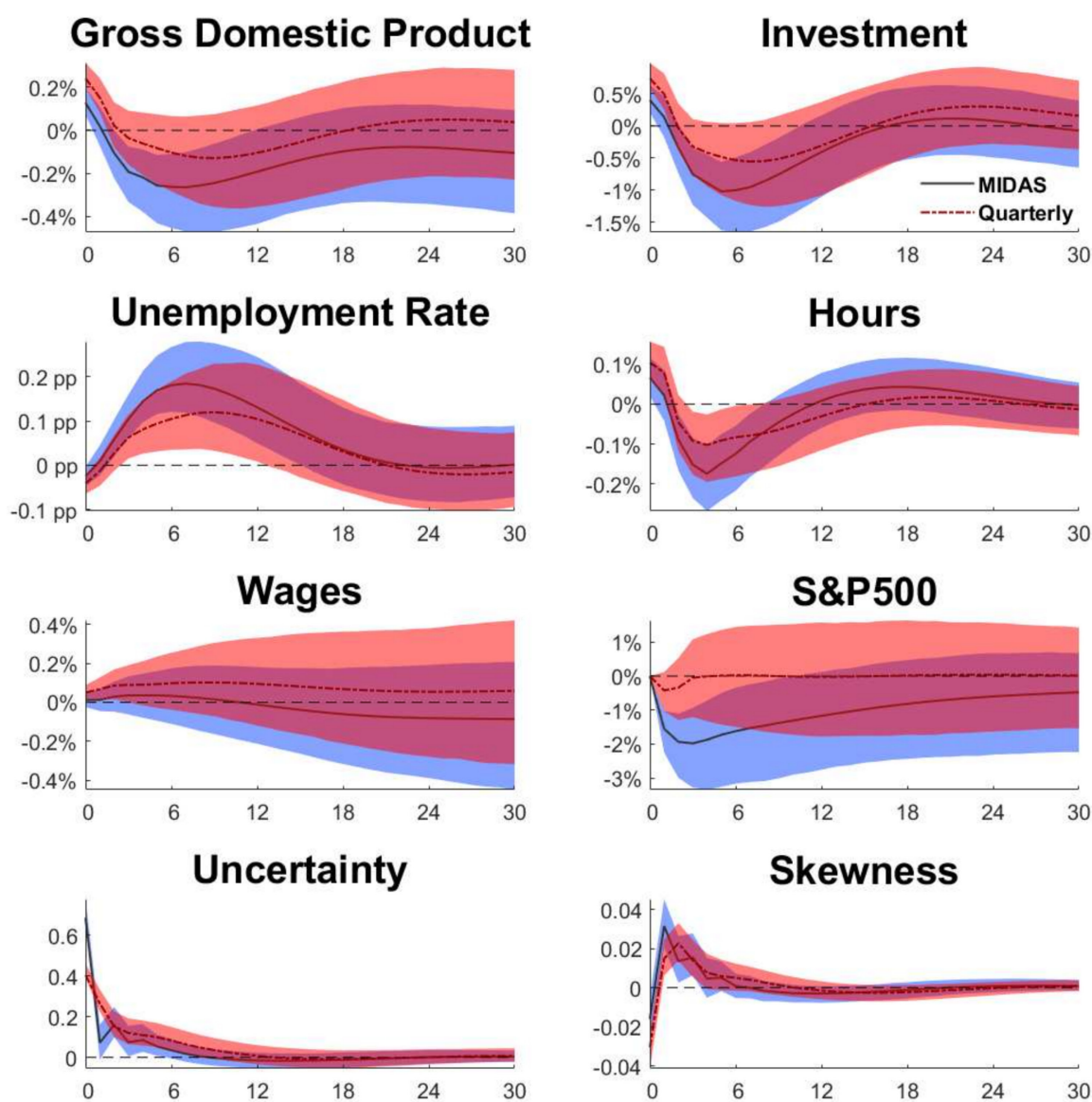


FIGURE 5 | Impulse responses of selected variables to an unexpected change in uncertainty: VARs modeling measures of uncertainty and skewness estimated with the MIDAS vs. quarterly-only models. Blue (solid) lines: MIDAS. Red (dashed) lines: Quarterly data-only. Shaded regions: 90% posterior credible sets. Unit variance shocks.

5 | VAR Analysis

Equipped with our measures of uncertainty and skewness, we run a VAR analysis to investigate the role that unexpected variations in these two measures may have played in shaping the US business cycle. We do so by modeling the following variables for the period 1971Q1–2022Q4:

$$X_t = \begin{bmatrix} 100\log(S\&P500) \\ \text{Uncertainty} \\ \text{Skewness} \\ 100\log(Wages) \\ 100\log(CPI) \\ 100\log(Hours) \\ \text{Unemployment rate} \\ 100\log(real\ GDP) \\ 100\log(real\ Investment) \\ \text{Shadowrate} \end{bmatrix}$$

This set of modeled variables is standard in the literature (see, e.g., Bloom 2009; Jurado, Ludvigson, and Ng 2015).¹⁷ Given the presence of the zero lower bound in our sample, we model the shadow rate proposed by Wu and Xia (2016) instead of the federal funds rate.¹⁸ The presence of investment in the VAR is justified by the attention posed by literature on the relationship between uncertainty and capital formation under non-convex adjustment costs (Bloom 2009; Bloom et al. 2018).¹⁹ On top of the usual macroeconomic indicators of the business cycle, we also model the novel measures of uncertainty and skewness obtained with our MIDAS approach.

We orthogonalize the residuals of the reduced-form VAR by assuming a recursive structure. We order uncertainty and skewness in the vector (and place them before the macroeconomic block) as in Salgado, Guvenen, and Bloom (2023). However, notice that no

TABLE 3 | Forecast error variance decomposition to uncertainty shocks using MIDAS vs. quarterly-only-based measures of uncertainty and skewness.

Indicator	MIDAS	Quarterly
GDP	0.08	0.03
Investment	0.10	0.04
Unemployment rate	0.18	0.07
Hours	0.12	0.07
Wages	0.00	0.03
S&P500	0.07	0.00
Uncertainty	0.80	0.66
Skewness	0.12	0.18

Note: Forecast error horizon: 2 years. Forecast errors are computed with two alternative VAR models, one modeling uncertainty and skewness measures estimated with our MIDAS predictive framework and the other one modeling uncertainty and skewness measures obtained with a quarterly-only predictive model.

particular economic meaning is attributed to the relative position of uncertainty and skewness in our vector, although it is worth noting that swapping their position leads to similar results. To ease interpretation, we model the measure of skewness in the VAR with its sign flipped around so that a rise in skewness identifies an increase in dispersion of the *left* tail of the real GDP distribution relative to the *right* tail (i.e., left skewness). Our Bayesian VAR features four lags. The informativeness of the Normal-Inverse Wishart prior distribution on the coefficients is set following Giannone, Lenza, and Primiceri (2015). Given that our sample comprises COVID-19 observations, we allow for temporary increases in volatility in the VAR residuals as in Lenza and Primiceri (2022).²⁰

5.1 | Uncertainty, Skewness, and the Business Cycle: Evidence

Figure 4 documents the impulse responses to a one-standard-deviation unexpected increase in uncertainty. Such responses are similar to those typically proposed by the literature; i.e., an increase in uncertainty is correlated with a temporary stock market bust, a decline in real activity (real GDP, hours, investment) and an increase in the unemployment rate.²¹ Prices react positively, something which can be interpreted with firms' incentives to set prices higher than in the absence of uncertainty as an insurance against future states characterized by high demand and low and sticky prices, an incentive known as “upward pricing bias” (see Fernández-Villaverde et al. 2015; Born and Pfeifer 2021). The policy rate is documented to increase in the short-run, a behavior consistent with a systematic monetary policy reaction to an inflation rate above the target. These responses are statistically significant and long-lasting, with real GDP, hours, and unemployment going back to their pre-shock level after about 3 years. A novel result is the strong response of skewness (i.e., a deterioration of the left tail of GDP growth distribution that is stronger than the expansion of the right tail) which suggests that uncertainty and skewness can hardly be

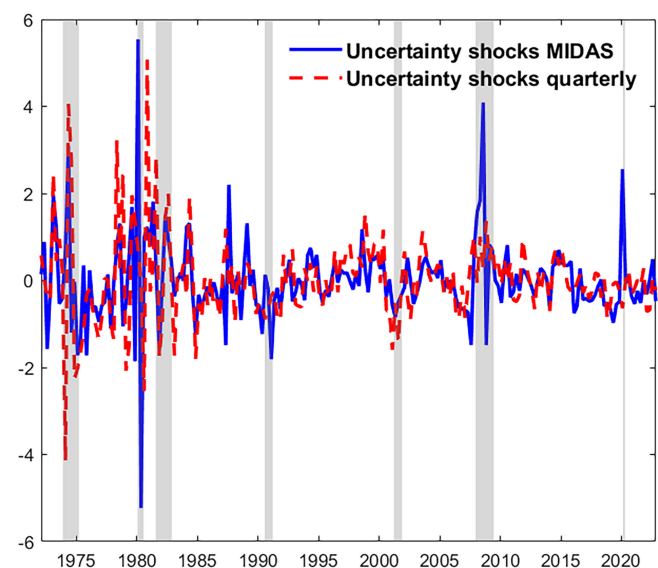


FIGURE 6 | Unexpected Changes in uncertainty: MIDAS vs. quarterly-data-only models. Unexpected changes in uncertainty in VARs including uncertainty and skewness measures computed with quarterly data-only vs. the MIDAS model.

thought of as independent processes. This finding is consistent with the empirical results documented by Hengge (2019) and Jovanovic and Ma (2022), who find that measures of uncertainty can significantly predict the lower tail of real GDP growth.

5.2 | Uncertainty, Skewness, and the Business Cycle: Frequency Matters

The impulse responses documented above are conditional on a VAR estimated with uncertainty and skewness measures obtained with our MIDAS framework. But how relevant is it to account for monthly observations of financial conditions when it comes to constructing measures of uncertainty and skewness with the ultimate goal of understanding their conditional correlations with the business cycle? In other words, how would our impulse responses and variance decomposition analysis look like if we instead employed measures of uncertainty and

skewness constructed with a quarterly data-only approach? To address these questions, we rerun our VAR analysis by replacing our measures of uncertainty and skewness with those obtained by working with the conditional density of real GDP growth estimated only with quarterly observations.

Figure 5 contrasts our baseline impulse responses with those obtained with measures of uncertainty and skewness based on quantiles estimated with quarterly data only. Such a figure, which focuses on key selected variables often studied by the uncertainty literature such as real GDP, investment, unemployment, hours, wages, the S&P500 index, as well as skewness, reveals that impulse responses produced with measures of uncertainty and skewness obtained with quarterly data only approach tend to underestimate the role of uncertainty innovations for the business cycle, with the peak responses of real activity indicators being substantially dampened with respect to the baseline ones.

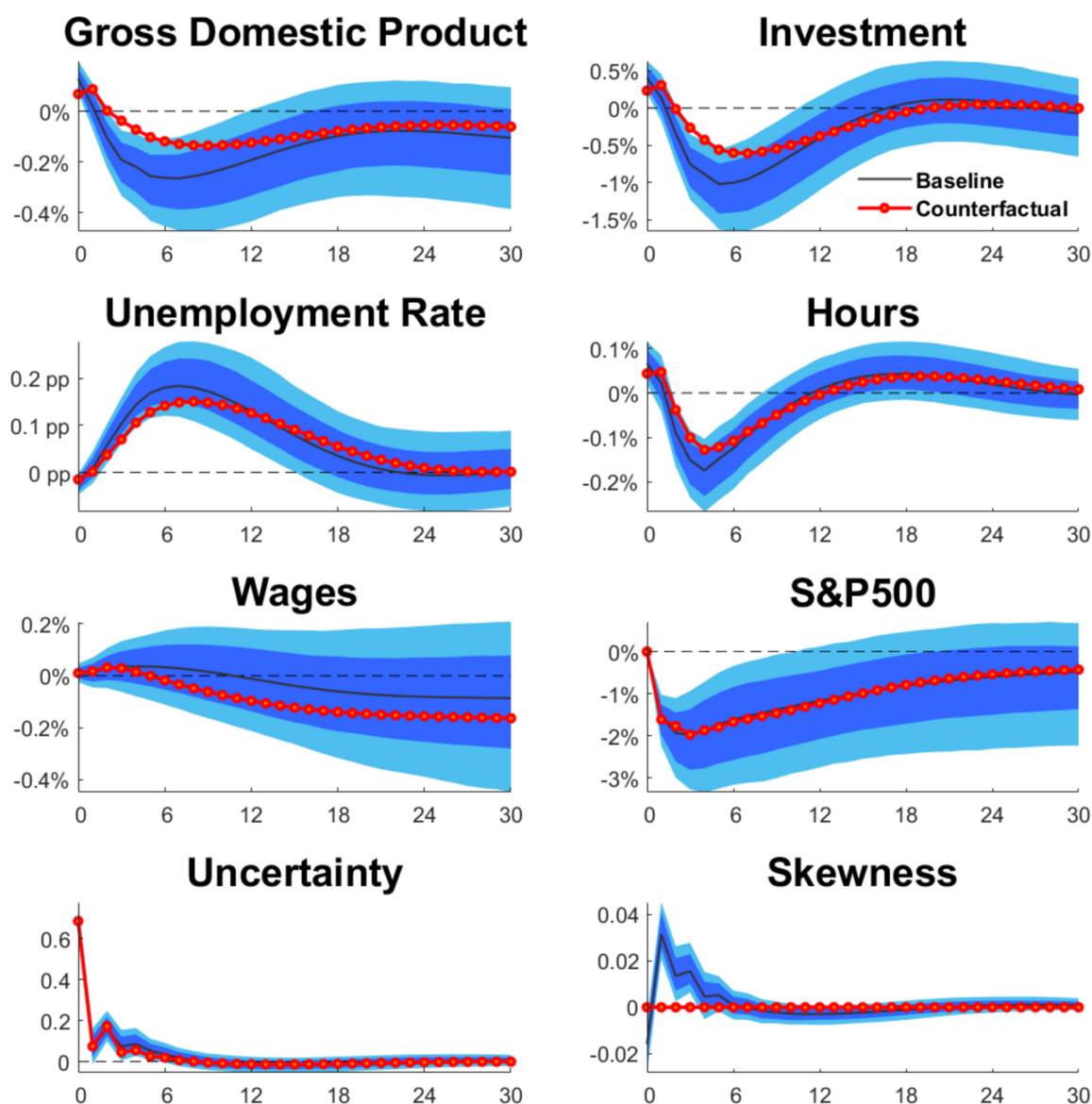


FIGURE 7 | Counterfactual reaction of real activity to an uncertainty shock: role of the endogenous response of skewness. The counterfactual responses are characterized by a muted reaction of skewness to an uncertainty shock. Skewness is counterfactually kept at zero at all horizons via fictitious shocks. Shaded regions: 68% and 90% posterior credible sets.

The forecast error variance decomposition analysis documented in Table 3 confirms the biases implied by the use of measures of uncertainty and skewness computed via quantile regressions relying on financial quarterly data only. Just 7% of the 2-year ahead forecast error volatility of the unemployment rate and 3%–4% of real GDP and investment are associated with unexpected changes in uncertainty. Differently, the version of the VAR that models our MIDAS measures of uncertainty and skewness points to contributions of uncertainty shocks as large as 18% for unemployment, 10% for investment, and 8% for real GDP.

What is the source of the discrepancies between impulse responses documented above? Figure 6 plots the series of unexpected changes in uncertainty according to the two models we are contrasting, i.e., the MIDAS model and the quarterly data-only framework. As one can easily see, using quarterly data only has got severe implications when it comes to correctly quantifying the size of such changes in crucial moments of the US economic history. Large recessions—in particular, the Great Recession and the COVID-19 one—are associated with large positive changes in uncertainty according to the MIDAS model, but much milder changes according to the quarterly data-only

framework. This evidence offers a *rationale* for the discrepant impulse responses documented in Figure 5, i.e., the quarterly data-only model underestimates the size of the unexpected positive changes in uncertainty and, therefore, the impact of such changes on real activity.

Wrapping up, our main finding in this section is that omitting monthly information on financial conditions when predicting the conditional density of real GDP growth has implications not only for the estimated conditional densities *per se* but also for the moments implied by such densities.

5.3 | The Uncertainty-Skewness Multiplier

It is of interest to understand how large a role the endogenous response of skewness plays in transmitting—and possibly magnifying—uncertainty shocks to real activity. We address this question by running a counterfactual exercise with our VAR in which we shut down the response of skewness and compare the factual and counterfactual impulse responses to an uncertainty shock.²²

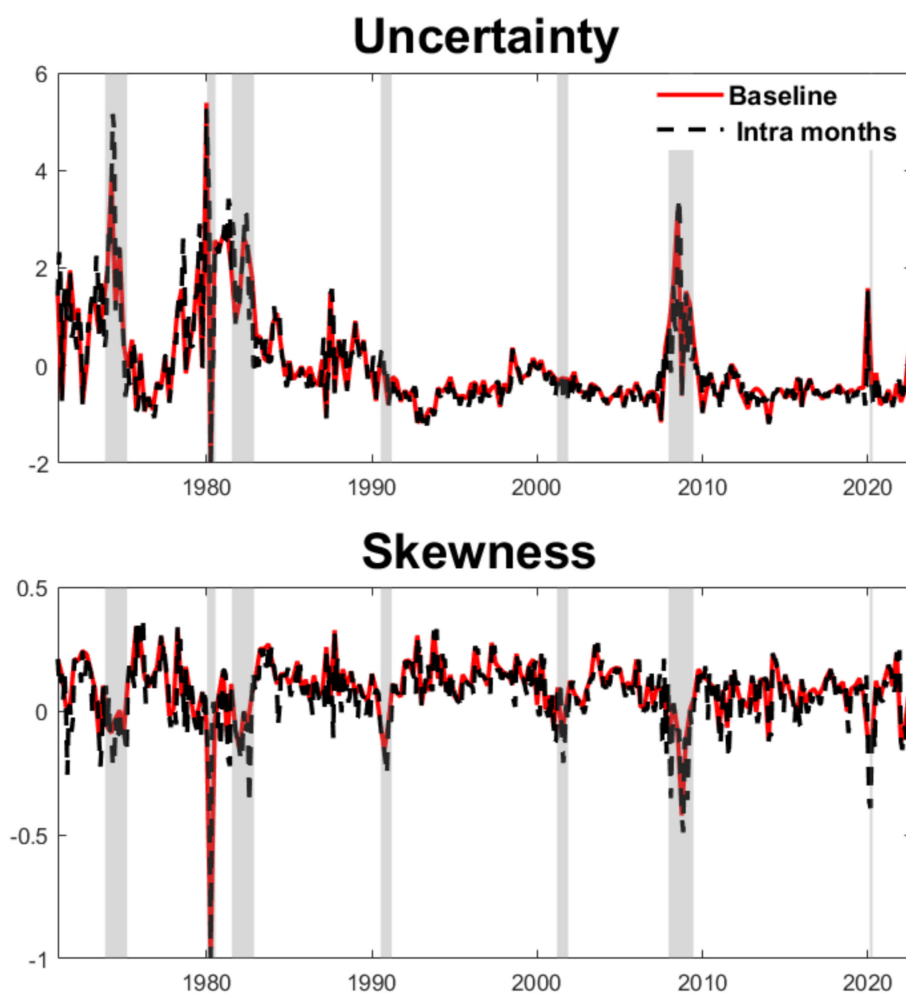


FIGURE 8 | Uncertainty vs. skewness indices: baseline and intra-months specification. The baseline measures are constructed by considering a MIDAS predictive framework that relies on information available up to the quarter $t - 1$ to model the conditional density of quarterly real GDP growth at quarter t . Differently, the intramonths specification allows monthly information on financial conditions available in quarter t to “nowcast” the conditional density of quarterly real GDP growth in the very same quarter.

Figure 7 documents the outcome of this exercise.²³ The responses computed under the counterfactual assumption of skewness not responding to uncertainty shocks point to a less severe recession and a more gradual return to the pre-shock paths for GDP, investment, and unemployment. The “uncertainty-skewness” multiplier (assessed by considering the peak responses of these three variables) is large: The response of skewness amplifies the impact of the uncertainty shock on real GDP, investment, and unemployment of about 95%, 68%, and 22%, respectively. Differently, the multiplier effect on hours and wages is fairly modest.

Our Appendix (Figure S8) documents the robustness of our results to the following perturbations of the baseline exercise: (i) changes in lag specification, (ii) different orderings, and (iii) an alternative scheme that separately identifies uncertainty and

skewness “shocks” through the max-share approach proposed by Uhlig (2003).²⁴

6 | A Monthly Frequency Analysis

Our approach has so far exploited a combination of monthly and quarterly data to model the one-quarter-ahead conditional density of the US real GDP growth. However, the monthly availability of the NFCI enables us to update our information set as new monthly information comes by. Conditional on the realization of real GDP growth at quarter t , we can model the conditional density of real GDP growth at quarter $t + 1$ by using NFCI realizations of the last month of quarter t (that is, what we have done so far) and also of the first month of quarter $t + 1$ and then the second month of quarter $t + 1$.²⁵ Via this “now-casting” procedure,

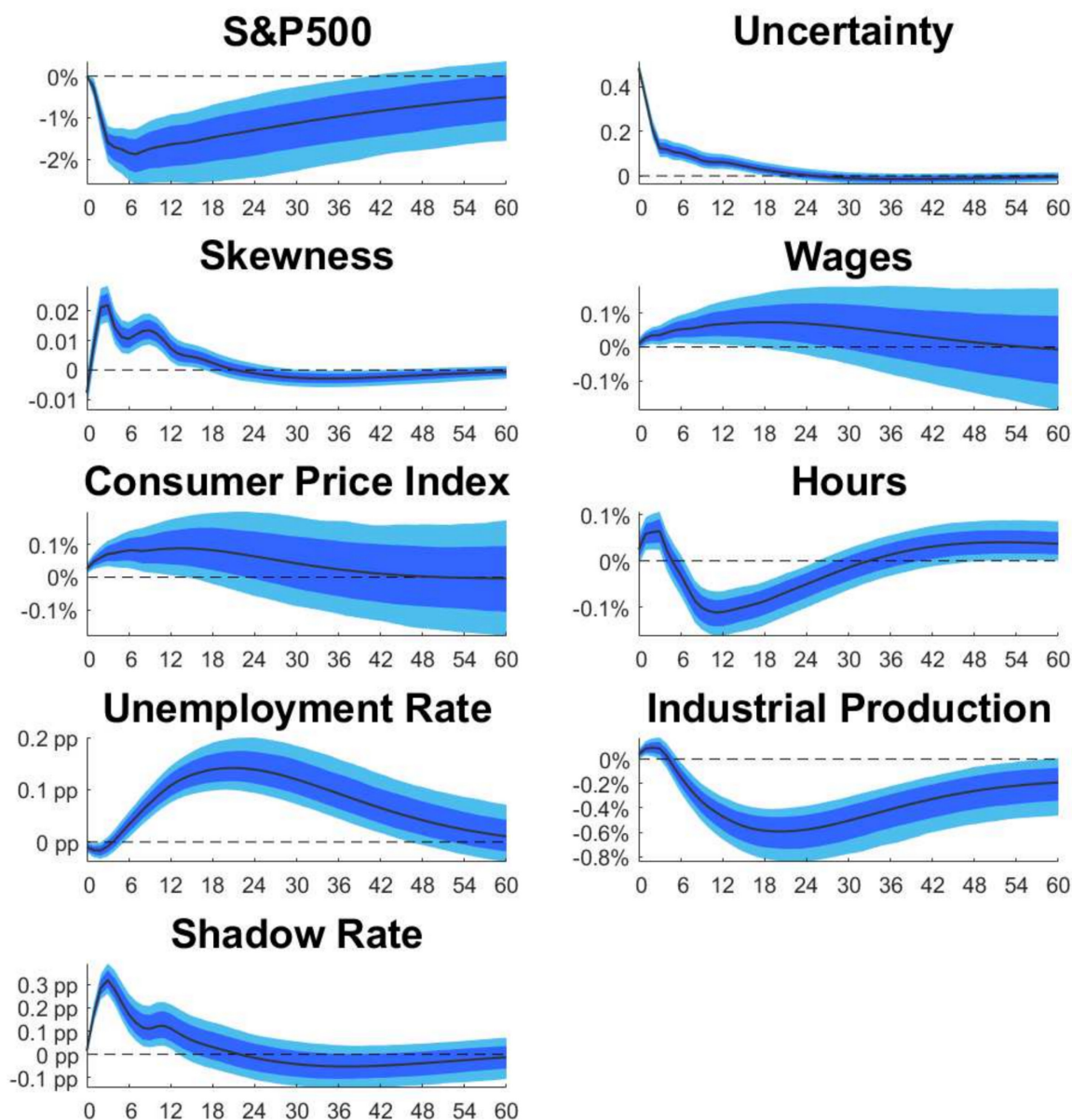


FIGURE 9 | Impulse responses to an unexpected change in uncertainty: Uncertainty and skewness measures computed at a monthly frequency with the MIDAS model. Size of the change: One standard deviation. VAR model featuring 12 lags. Shaded regions: 68% and 90% posterior credible sets.

we can construct measures of uncertainty and skewness at a monthly frequency.

Figure 8 displays the so-obtained measures and compares them to those obtained with the baseline approach. Our monthly measures are as follows: (i) highly correlated with the quarterly ones, (ii) slightly more volatile, and (iii) capture large (in absolute terms) realizations of uncertainty and skewness that are (naturally) smoothed out by the corresponding quarterly representations. Importantly, these higher frequency measures of uncertainty and skewness enable us to repeat our VAR exercise at a monthly frequency, which we do by replacing real GDP (unavailable monthly) with industrial production and by dropping investment (again, unavailable monthly). Given the change in frequency, we estimate a Bayesian VAR with 12 lags. We keep using the same prior specification and correct for COVID outliers.

Figure 9 shows that the empirical findings proposed in Section 4 and obtained with our quarterly proxies are robust to moving to a monthly analysis, i.e., uncertainty shocks are recessionary, inflationary, increase skewness (the left tail of real GDP growth), and are associated with a monetary policy tightening. Our Appendix (Figure S9) documents that the role of skewness as a “magnifier” of the business cycle effects of uncertainty shocks is confirmed when working with monthly data.

One advantage of working with this higher frequency is that of being able to directly compare our uncertainty proxy with those proposed by the existing literature. In particular, Ludvigson, Ma, and Ng (2021) work with the same forecasting approach proposed by Jurado, Ludvigson, and Ng (2015) and produce monthly proxies for financial and macroeconomic uncertainty, while Carriero, Clark, and Marcellino (2018) and Shin and Zhong (2020) propose sophisticated multivariate models to estimate macroeconomic

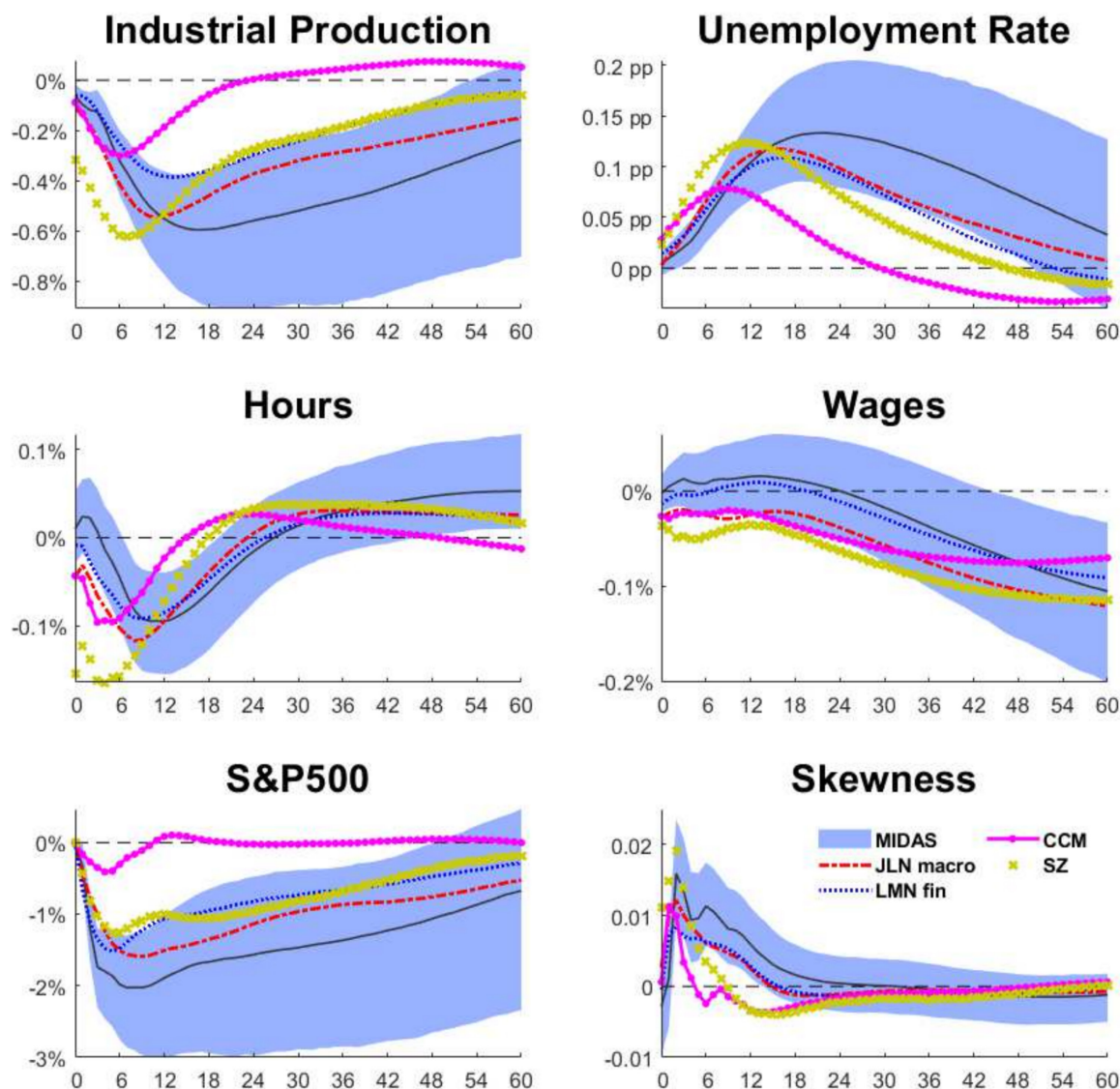


FIGURE 10 | Impulse responses of selected variables to an unexpected change in uncertainty: VARs embedding measures of uncertainty estimated with the MIDAS model vs. alternative monthly measures available in the literature. Alternative measures: macroeconomic and financial uncertainty by Jurado, Ludvigson, and Ng (2015) (JLN macro) and Ludvigson, Ma, and Ng (2021) (LMN fin); uncertainty by Carriero, Clark, and Marcellino (2019) (CCM); uncertainty by Shin and Zhong (2020) (SZ). Shaded regions: 90% posterior credible sets. Impulse-response functions are computed using monthly VARs featuring the same variables and 12 lags. VARs are estimated on the largest common sample in which all measures are available (1985M1–2012M12).

uncertainty. We then exploit the possibility of closely comparing our proxy for uncertainty with those in the existing literature by replacing our proxy for uncertainty with those in the literature (one at a time) in our monthly VAR. To enhance comparability, we present results based on a common sample, i.e., 1985M1–2012M12. The beginning of the sample is dictated by the availability of the Carriero, Clark, and Marcellino (2018) proxy, while the end by the one by Shin and Zhong (2020).

Figure 10 plots the impulse responses estimated by replacing our monthly proxy for uncertainty with those produced by the authors cited above (included in the VAR one at a time). A few comments are in order. First, each of these measures of uncertainty is associated with a recession. Second, each one is associated with a short-run and short-lived response of skewness. Third, our measure of uncertainty implies a longer-lived response of industrial production and unemployment, with peaks that are similar to, if not larger than, those generated by the other measures of uncertainty. Fourth, the response of hours and wages associated with our measures of uncertainty is weaker than those generated by these alternative proxies (in particular, than the one induced by the Shin and Zhong (2020) proxy). Unexpected movements in uncertainty associated with our proxy are at least comparable to, if not more powerful than, those associated with the other proxies proposed by the literature as far as industrial production, unemployment rate, and skewness are concerned.²⁶ Moreover, our proxy for uncertainty is the only one constructed consistently with our proxy for skewness in our analysis. Hence, given that accounting for the endogenous response of skewness is important for the role of uncertainty shocks to be correctly pinned down, our measure of uncertainty presents the advantage of being consistent—in terms of construction—with our measure of skewness.

7 | Conclusions

This paper shows that employing monthly frequencies of financial conditions improves the predictive power of an otherwise quarterly data-only quantile regressions framework when it comes to modeling the conditional density of real GDP growth. We support this statement with a battery of metrics and statistical tests. Then, we use our mixed frequency quantile regression approach to construct novel measures of uncertainty and skewness, which are obtained by combining selected estimated quantiles. Equipped with these novel measures, we run VAR analysis to quantify the relevance of unexpected changes in uncertainty and skewness for the business cycle. We find such changes to be statistically and economically connected to variations in real activity. A different picture emerges when computing measures of uncertainty and skewness based on a time-varying conditional density of real GDP growth estimated without taking information on financial conditions available at a monthly frequency. In particular, the contribution of uncertainty shocks as drivers of the business cycle is underestimated. A counterfactual exercise computed by shutting down the endogenous response of skewness to an uncertainty shock reveals the existence of an “uncertainty-skewness multiplier”, i.e., endogenous skewness acts as an amplifier of the real effects of uncertainty shocks. Finally, we construct monthly counterparts of our novel uncertainty and skewness measures and document the robustness of our baseline results to working with higher frequencies. This

last VAR exercise also enables us to compare more closely the business cycle implications of unexpected increases in uncertainty as proxies by our novel measures with those conditional on other proxies recently proposed by the literature. We find the effects of our proxy as being no weaker (if anything, stronger) than those associated with the existing measures.

Our results suggest that early-warning models on vulnerable real GDP growth should embed financial indicators available monthly to better quantify tail risks, uncertainty, and their empirical links with the business cycle. From a theoretical modeling standpoint, our evidence points to the need of building up frameworks featuring mechanisms that generate the type of uncertainty-skewness interaction we find in the data. Recent proposals include downward wage rigidities (Cacciatore and Ravenna 2021), the zero lower bound (Caggiano, Castelnuovo, and Pellegrino 2017; Basu and Bundick 2017), state-dependent households' risk aversion (Caggiano, Castelnuovo, and Pellegrino 2021; Bretscher, Hsu, and Tamoni 2023; Pellegrino, Castelnuovo, and Caggiano 2023), firms' leverage (Jensen et al. 2020), firms' nominal upward pricing bias (Andreasen et al. 2024), and the rapid adoption of new technologies (Jovanovic and Ma 2022).

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Data Availability Statement

The dataset and estimation codes to replicate all the results in the paper can be downloaded from the JAE's data archive.

Open Research Badges



This article has been awarded Open Data Badge for making publicly available the digitally-shareable data necessary to reproduce the reported results. Data is available at Data DOI: [10.15456/jae.2024248.0759770713](https://doi.org/10.15456/jae.2024248.0759770713).

Endnotes

¹ As pointed out by Stock and Watson (1999) (p. 15), “[...] the cyclical component of real GDP is a useful proxy for the overall business cycle”.

² For the seminal paper on quantile regressions, see Koenker and Bassett (1978). An analysis of the subsequent methodological advances made in the quantile regression areas is offered by Koenker (2005). An extensive presentation and investigation of the MIDAS approach can be found in Ghysels, Santa-Clara, and Valkanov (2004) and Ghysels, Sinko, and Valkanov (2007).

- ³ For instance, to predict the conditional density of real GDP in 2019Q4, we use information of NFCI at a monthly level first until September 2019, then update the information set by including the October 2019 realization of NFCI and finally the November 2019 one. This is in contrast with the standard predictive exercise where monthly information of a given quarter dated $t-1$ is employed to model the conditional density of real GDP of the following quarter t .
- ⁴ A broader discussion of the connections between our paper and the literature on uncertainty, skewness, and the business cycle can be found in Castelnuevo and Mori (2022) (the working paper version of this paper).
- ⁵ Rossi and Sekhposyan (2017) apply this methodology to Euro area aggregate and country-specific data.
- ⁶ The NFCI captures the evolution of broad financial conditions based on information on money markets, debt, and equity markets, and the traditional and shadow banking systems.
- ⁷ Real-time and aggregation issues regarding financial conditions have also been investigated in the forecasting and uncertainty realms. Amburgey and McCracken (2023b) evaluate the real-time predictive content of NFCI vintages and find them to be informative. Related papers are Forni et al. (2021) and Loria, Matthes, and Zhang (2022), who show that identified financial and monetary policy shocks are relevant drivers of macroeconomic risk. Paccagnini and Parla (2021) show that VAR investigations on the impact of financial uncertainty shocks conducted with low-frequency data may be affected by a temporal aggregation bias, an issue that can be tackled by augmenting the information set with higher frequency financial information.
- ⁸ $NFCI_{t,mi}$ with $i \in \{1, 2, 3\}$ is the observation of NFCI at the i -th month of quarter t .
- ⁹ The NFCI is available weekly. Carriero, Clark, and Marcellino (2022) explore the gains of employing weekly indicators when conducting nowcasting exercises, and find them to be mild and not necessarily significant compared to using monthly realizations. The choice of using monthly-level regressors (as opposed to weekly observations) enables us to work with a parsimonious - yet powerful - predictive model.
- ¹⁰ In short, we assume the real GDP growth series to follow an AR(1) process and allow for jumps in the volatility of the process for the first three quarters of 2020 (given that real GDP start collapsing in 2020Q1), with a decay rate applied from 2020Q4 onward. We then estimate our quantile regression model by working with the median (across draws) of the real GDP fitted series rescaled by using (draw-specific) volatility scaling and decay parameters.
- ¹¹ The choice of the regressors in our MIDAS framework represents a natural extension of the quarterly-only model proposed by Adrian, Boyarchenko, and Giannone (2019). The limited number of regressors of our framework is also intended to tackle the parameter proliferation problem often arising when dealing with quantile regressions with mixed-frequency data. For a formal algorithm that deals with this issue, see Lima, Meng, and Godeiro (2020).
- ¹² We see our choice of focusing on a one quarter-ahead forecasting analysis as a natural consequence of dealing with monthly-level information of financial activity - that evolves quite dynamically - to predict the future conditional density of real GDP growth. Our Appendix (Figure S3) documents that, when working with a $h=1$ -year forecasting horizon, the MIDAS approach clearly helps capture the left-tail of the real GDP growth distribution while it does not contribute significantly to modeling the right tail. Since we will use the information on both tails to construct measures of uncertainty and skewness, we prefer to focus on the horizon $h=1$.
- ¹³ A similar correction materialized just prior to the acceleration of the Great Recession. At that time, the monthly NFCI registered a lower value in August 2008 with respect to July 2008, to then substantially increase when moving from August to September 2008.
- ¹⁴ $\overline{QS}_{MIDAS}$ and $\overline{QS}_Q$ denote the sample average of the quantile scores in the forecasting period considered. The same holds true for $\overline{qwCRPS}_{MIDAS}$ and $\overline{qwCRPS}_Q$.
- ¹⁵ The goodness of the calibration of the predictive density is also supported by a PIT-based test (documented in our Appendix, Figure S5). The empirical cumulative distribution of the MIDAS model is well-calibrated at the tails and overall performs as the quarterly-only model.
- ¹⁶ The distinction between financial and macroeconomic uncertainty is relevant when it comes to isolating uncertainty shocks. According to Ludvigson, Ma, and Ng (2021), financial uncertainty shocks (as opposed to macroeconomic uncertainty shocks) are likely to be drivers of the US business cycle. Differently, Carriero, Clark, and Marcellino (2019) and Forni, Gambetti, and Sala (2021b) point to macroeconomic uncertainty as a relevant driver of the business cycle. For related papers pointing to both types of shocks as possible drivers of the US business cycle, see Angelini and Fanelli (2019), Angelini et al. (2019), and Shin and Zhong (2020).
- ¹⁷ Bloom (2009) considers employment, while we model the unemployment rate because the latter is often at the center of debates regarding the labor market and the stance of the business cycle. The correlation between the yearly growth rate of nonfarm employment and the unemployment rate in our sample is -0.41 . Our results are robust to replacing the unemployment rate with employment. The exact definition of the variables we model can be found in our Appendix.
- ¹⁸ Our results are robust to the employment of the alternative measure of the shadow rate proposed by Krippner (2020).
- ¹⁹ Following Justiniano, Primiceri, and Tambalotti (2011), we define investment as the sum of personal consumption expenditures on durables and gross private domestic investment. Our results are robust to considering investment-only data in the VAR.
- ²⁰ The way in which we handle COVID-19 observations does not drive our results. We show in the Appendix (Figure S6) that we obtain similar results when (i) estimating the VAR with standard methods that do not correct for extreme observations and (ii) we focus on a sample that ends in 2019.
- ²¹ Notice that we find a *short-run* drop in skewness and increase in real GDP growth after an uncertainty shock. While counterintuitive, these reactions are short-lived and quickly revert to more conventional signs. Moreover, our Appendix (Figure S6) shows that this evidence is sample-dependent and disappears when working with the 1985–2022 sample.
- ²² Our counterfactual scenario is constructed by subjecting the economy to fictitious shocks to skewness that fully counteract the effects of the uncertainty shock on skewness (Sims and Zha 2006). For references on the use of fictitious shocks in counterfactual VAR investigations, see Bernanke, Gertler, and Watson (1997), Hamilton and Herrera (2004), and Kilian and Lewis (2011). A discussion on this way of running counterfactual simulations vs. alternative ones can be found in Antolín-Díaz, Petrella, and Rubio-Ramírez (2021) and McKay and Wolf (2023).
- ²³ The negligible impact on the dynamics of skewness on the stock market is largely due to the zero restriction related to the contemporaneous effect of the former on the latter. When ordering uncertainty and skewness before the S&P500 index, we find a milder response of the S&P500 index to an uncertainty shock in the counterfactual world in which the endogenous response of skewness is muted (see Figure S7). The counterfactual responses of the remaining variables are essentially unaffected by this change.
- ²⁴ Identification is achieved by requiring the uncertainty shock to generate the largest response (increase) in uncertainty for the first year after the shock. The skewness shock (which we need to run our counterfactual simulation) is identified as the innovation that: (i) is

orthogonal to the uncertainty disturbance; and (ii) generates the largest increase in skewness for the first year. For a similar identification strategy to disentangle uncertainty and financial shocks, see Caldara et al. (2016).

²⁵ Note that we use three realizations of NFCI to generate each given monthly realization, i.e., we employ a fixed number of regressors each time we predict the conditional density of real GDP.

²⁶ Table S2 in our Appendix documents the outcome of a 2-year ahead forecast error variance decomposition analysis, which offers support to this statement.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.